1.

$$y = C - \frac{2\pi^2 k}{mg}(x-a)^2 \quad (C$$

2.

$$\frac{7}{12}R\ .$$

3.

$30\ \mu\mathrm{m}$.

4.

$U_{\mathrm{O}} = \dfrac{kT}{e}\ln\left(\dfrac{I_{\mathrm{SC}}}{I_{\mathrm{S}}}+1\right)$ e k T

I_{S} p-n

1 $\boldsymbol{r}' = \alpha\boldsymbol{r}$ 、 $t' = \beta t$

$$\boldsymbol{v}' = \lim_{\Delta t'\to 0}\frac{\Delta \boldsymbol{r}'}{\Delta t'} = \frac{\alpha}{\beta}\lim_{\Delta t\to 0}\frac{\Delta \boldsymbol{r}}{\Delta t} = \frac{\alpha}{\beta}\boldsymbol{v}\ , \tag{1}$$

$$\boldsymbol{a}' = \lim_{\Delta t'\to 0}\frac{\Delta \boldsymbol{v}'}{\Delta t'} = \frac{\alpha}{\beta^2}\lim_{\Delta t\to 0}\frac{\Delta \boldsymbol{v}}{\Delta t} = \frac{\alpha}{\beta^2}\boldsymbol{a}\ . \tag{2}$$

$$m\boldsymbol{a} = \boldsymbol{F}(\boldsymbol{r})\ , \tag{3}$$

$$m\boldsymbol{a}' = \boldsymbol{F}(\boldsymbol{r}')\ . \tag{4}$$

(2) $\boldsymbol{F}(\alpha\boldsymbol{r}) = \alpha^k\boldsymbol{F}(\boldsymbol{r})$, (4)

$$m\boldsymbol{a} = \alpha^{k-1}\beta^2\boldsymbol{F}(\boldsymbol{r})\ . \tag{5}$$

(3)

$$\beta = \alpha^{\frac{1-k}{2}}\ . \tag{6}$$

2 $k = -2$

$$l' = \alpha l \tag{7}$$